

CubeSat Attitude Determination and Control System Hardware-in-the-Loop Validation Using a 3-DOF Air Bearing Project Proposal

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1 Objective

The objective of this project is to classify performance metrics and system objectives for a three-axis CubeSat Attitude Determination and Control System (ADCS). The system will be evaluated through a hardware-in-the-loop (HIL) testbed mounted on a three-degree-of-freedom air bearing representative of space conditions.

2 System Overview

A flight-representative CubeSat platform will be mounted on a 3-DOF air bearing to enable near-frictionless rotation. Attitude actuation will be provided by a set of reaction wheels, while onboard sensors provide state feedback. A microcontroller executes the ADCS algorithms in real time, interfacing with both physical hardware and a dynamics simulator.

3 ADCS Architecture

Attitude estimation is performed using sensor fusion techniques to estimate spacecraft attitude and angular rate. Control laws generate reaction wheel torque commands to perform detumbling, orientation transition (slew maneuvers), and steady-state pointing. The controller is designed to respect actuator limits while maintaining stability and performance.

4 Hardware-in-the-Loop Framework

The HIL framework couples the physical spacecraft-representative hardware with a simulated orbital environment. Reaction wheel torques are applied to the air-bearing platform, while environmental effects and sensor noise are injected through simulation. This enables validation of control performance and timing under realistic operating conditions.

5 Performance Metrics

The system will be evaluated using quantitative metrics including:

- Detumbling to angular rates below ? deg/s
- Detumbling time below ? s
- Steady-state pointing error below ? deg
- Slew maneuver settling time under ? s
- Real-time firmware performance

6 Expected Outcomes

This project will deliver a fully integrated CubeSat ADCS HIL testbed, experimental validation of closed-loop real-time attitude control, quantified detumbling/pointing/slew maneuver demonstrations, and documented lessons learned applicable to flight CubeSat missions.

7 Milestones

- Literature review and requirements definition
- Control law derivation and simulation
- Embedded control system framework implementation
- Mechanical modeling and fabrication
- Attitude estimator and control law implementation
- Hardware-in-the-loop and 3D bearing integration
- Experimental validation and final report